Deeper examination of some functions of the `datana` package

Examples

April 7, 2026

This document expands on examples of different functions from the datana package (Salas-Eljatib and Campos 2025), implemented in R (R Core Team 2025). The main idea of such a report is to provide most of the utilities and versatility of the aforementioned package's functions.

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1 interp

1.1 About this function

The `interp` function was created for the analysis process of taper data. The idea is that with this function missing data is generated (*i.e.* commercial heights and diameters) which allow to obtain commercial volumes, in addition to be a requirement in order to the data to complain to the following assumptions:

1. Decreasing diameter in height.
2. To always exists a section at stump height (generally at 0.3 m), dbh height, and between the mentioned values (generally 0.8m).
3. That the diameter outside bark is always larger than the diameter inside bark for the same height.
4. That the last section (the tip of the tree) has diameter equals 0 (zero) and $hl = h$.

1.2 Example

Loading taper data of a tree:

```
df <- read.csv((./resources/ahusa.csv"))
head(df[, c("dl.cc", "hl")])
```

```
dl.cc   hl
1 11.02 0.05
2  8.46 1.30
3  8.10 1.58
4  7.86 2.37
5  6.90 3.16
6  6.70 3.95
```

This tree doesn't have a section at height 0.8 m, so this is interpolated.

```
library(datana)
## the height will be 0.8 m
new.section <- 0.8
interpolated <- interp(x = df$dl.cc, y = df$hl, ylu = new.section)
interpolated$datares
```

```
x    y interpolated
1  11.020000 0.05        FALSE
2   9.378939 0.80         TRUE
3   8.460000 1.30        FALSE
4   8.100000 1.58        FALSE
5   7.860000 2.37        FALSE
6   6.900000 3.16        FALSE
7   6.700000 3.95        FALSE
8   6.000000 4.74        FALSE
9   5.500000 5.53        FALSE
```

```
10  4.400000 5.60        FALSE
11  3.500000 6.32        FALSE
12  2.820000 7.11        FALSE
13  0.000000 9.00        FALSE
```

2 plotrend

2.1 Sobre esta función

2.2 Datos para el ejemplo

Se cargan los datos de ahusamiento de un árbol

```
df <- datana::maple
head(df)
```

```
tree  dbh leaf branch   bole  bark  total
1    1  9.1 0.67  21.30   9.94  1.51  33.42
2    2  9.1 0.54  20.30   6.21  0.63  27.68
3    3  9.9 0.72  19.26  13.93  1.69  35.60
4    4 10.2 0.60  21.77  17.68  1.60  41.65
5    5 16.7 2.90  28.47 101.94 11.37 144.68
6    6 17.2 2.39  26.94  90.26 12.35 131.94
```

Se ajusta un modelo lineal

```
m1 <- lm(leaf ~ dbh, data = df)
m1
```

```
Call:
lm(formula = leaf ~ dbh, data = df)

Coefficients:
(Intercept)          dbh
    -6.1498       0.5375
```

Usando la función

```
datana::plotrend(x = df$dbh, y = residuals(m1), xlab = "DBH (cm)")
abline(h = 0)
```

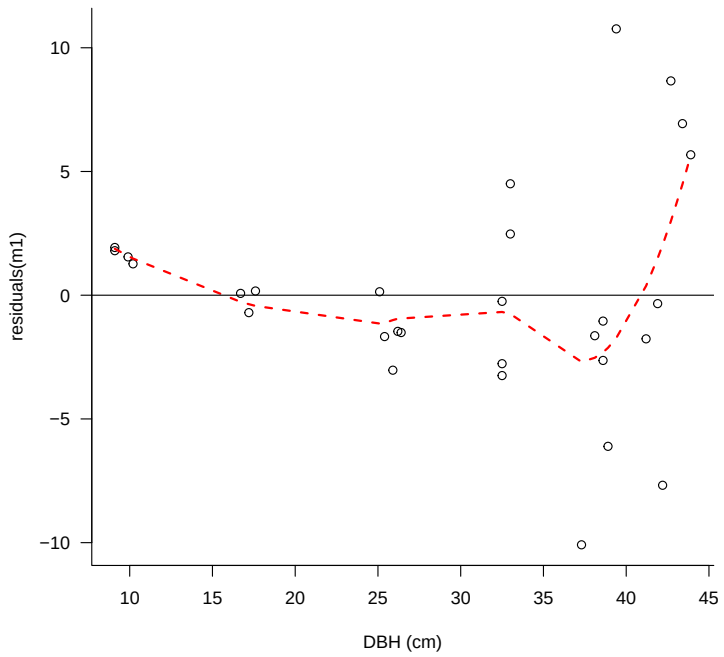

Figure 1: Example plot generated with the `plotrend` function.

References

R Core Team (2025). *R: A language and environment for statistical computing*. `http://www.R-project.org`. R Foundation for Statistical Computing. Vienna, Austria.

Salas-Eljatib, C. and N. Campos (2025). `datana`*: Data and functions to accompany Análisis de datos con R*. R package version 1.1.4. DOI: `10.32614/CRAN.package.datana`. URL: `https://cran.r-project.org/package=datana`.